

Exercise 1:

1. Read the article “A Big Data Modeling Methodology for Apache Cassandra”. Provide a ½ page summary including your comments and impressions.

- The article “A Big Data Modeling Methodology for Apache Cassandra” presents a well-organized approach for designing databases specifically for Cassandra, addressing challenges that traditional relational database modeling cannot fully solve. The authors explain that Cassandra is designed for large-scale systems, high availability, and fast performance. However, because it does not support joins, aggregations, and many common SQL operations, the usual relational strategy of “normalize first” is not effective. Instead, the paper recommends a query-focused modeling approach, where database tables are created based on the application’s query requirements so that data can be retrieved efficiently.

The methodology is built around four main ideas:

- **Understand the Data:** identify entities and relationships through a conceptual model that is independent of any specific technology.
- **Understand the Queries:** design tables around the application’s read patterns since queries determine the schema structure.
- **Data Nesting:** place related records together within the same partition to improve query speed and efficiency.
- **Data Duplication:** keep copies of the same information in multiple tables to eliminate the need for joins and support different query needs.

The paper also explains a set of mapping rules and design patterns that transform conceptual models into Cassandra-optimized table structures. In addition, it introduces Chebotko Diagrams, a visual representation that illustrates the relationships between tables, keys, and queries, making Cassandra schemas easier to understand. The authors further present a tool called KDM, which automates much of the modeling process, from conceptual design all the way to generating Cassandra Query Language (CQL) scripts.

Personal Comments & Impressions:

- Overall, the article makes Cassandra data modeling appear far more systematic and easier to understand rather than relying on trial and error. I appreciated how the authors highlighted the major differences between Cassandra and relational databases, particularly the idea that data duplication is expected and often necessary. The focus on designing around queries instead of beginning with the data itself felt like an important shift in thinking, but it becomes logical when considering Cassandra’s goal of serving requests with minimal disk access. I also found the Chebotko Diagrams especially useful because they provide a clear visual explanation of how Cassandra schemas are organized.
- In general, the article effectively explains why Cassandra is such a strong option for big-data applications while also emphasizing the importance of careful schema planning. The methodology seems valuable because it offers a clear, step-by-step framework that can help prevent poor schema design, which is a common issue for newcomers working with Cassandra.

Exercise 2:

- a.** Create and execute a file in your working (home) directory on the primary EMR node called `init.cql` and the `Execute ex2.cql`. Then execute the shell command `'DESCRIBE TABLE Music'`.

```
[cqlsh> source './init.cql';
[cqlsh> describe keyspaces;

a20581012  system_auth          system_schema  system_views
system     system_distributed      system_traces  system_virtual_schema

[cqlsh> USE a20581012;
cqlsh:a20581012> 
```

```
CREATE TABLE Music (
  artistName text,
  albumName text,
  numberOfSold int,
  cost int,
  PRIMARY KEY (artistName, albumName)
);
```

```
[cqlsh:a20581012> DESCRIBE TABLE Music;
CREATE TABLE a20581012.music (
  artistname text,
  albumname text,
  cost int,
  numbersold int,
  PRIMARY KEY (artistname, albumname)
) WITH CLUSTERING ORDER BY (albumname ASC)
AND additional_write_policy = '99p'
AND bloom_filter_fp_chance = 0.01
AND caching = {'keys': 'ALL', 'rows_per_partition': 'NONE'}
AND cdc = false
AND comment = ''
AND compaction = {'class': 'org.apache.cassandra.db.compaction.SizeTieredCompactionStrategy', 'max_threshold': '32', 'min_threshold': '4'}
AND compression = {'chunk_length_in_kb': '16', 'class': 'org.apache.cassandra.io.compress.LZ4Compressor'}
```

Exercise 3:

a. Execute ex3.cql. Provide (a) the content of the ex3.cql file as one result of this exercise, and (b) execute the command 'SELECT * FROM Music;' and provide a screenshot of the output of this command as another result of the exercise

```
INSERT INTO music (artistname, albumname, numbersold, cost)
VALUES ('Mozart', 'Greatest Hits', 100000, 10);
INSERT INTO music (artistname, albumname, numbersold, cost)
VALUES ('Taylor Swift', 'Fearless', 2300000, 15);
INSERT INTO music (artistname, albumname, numbersold, cost)
VALUES ('Black Sabbath', 'Paranoid', 534000, 12);
INSERT INTO music (artistname, albumname, numbersold, cost)
VALUES ('Katy Perry', 'Prism', 800000, 16);
INSERT INTO music (artistname, albumname, numbersold, cost)
VALUES ('Katy Perry', 'Teenage Dream', 750000, 14);

```

"ex3.cql" 14L, 563B 14,51 All

```
[cqlsh:a20581012> source './ex3.cql';
[cqlsh:a20581012> SELECT * FROM Music;
]
```

artistname	albumname	cost	numbersold
Mozart	Greatest Hits	10	100000
Black Sabbath	Paranoid	12	534000
Taylor Swift	Fearless	15	2300000
Katy Perry	Prism	16	800000
Katy Perry	Teenage Dream	14	750000

(5 rows)
cqlsh:a20581012> █

Exercise 4:

- a. Provide (a) the content of the ex4.cql file and (b) a screenshot of the output of executing this file as the result of this exercise

```
SELECT * FROM music WHERE artistname = 'Katy Perry';
```

"ex4.cql" 1L, 53B 1,1 All

```
[cqlsh:a20581012> source './ex4.cql';
```

artistname	albumname	cost	numbersold
Katy Perry	Prism	16	800000
Katy Perry	Teenage Dream	14	750000

```
(2 rows)  
cqlsh:a20581012>
```

Exercise 5:

a. Provide (a) the content of the ex5.cql file and (b) a screenshot of the output of executing this file as the result of this exercise.

```
SELECT * FROM music
WHERE numbersold >= 700000
ALLOW FILTERING
```

"ex5.cql" 3L, 66B 3, 16 All

```
[cqlsh:a20581012> source './ex5.cql';
]
```

artistname	albumname	cost	numbersold
Taylor Swift	Fearless	15	2300000
Katy Perry	Prism	16	800000
Katy Perry	Teenage Dream	14	750000

(3 rows)
cqlsh:a20581012>